

Agentic Healthcare Assistant

Nour Habib

Overview

UI: Streamlit front-end with two views: Patient and Doctor

- Patient view: 3 different patients can be selected from the drop-down menu, each of which have their own medical histories
 - Patient can also book/cancel/edit their own appointments
- Doctor view: shows 2 doctors can be selected from the drop-down menu
 - Doctor can search for patients from the SQL database
- Search bar: used for health-related questions by calling the Medline API

Backend: SQLite database

- I used SQLite to create tables for patients, doctors and appointments

AI backend: LangGraph, FAISS

- I made heavy use of LangGraph to create the chatbot and tools
- I built the RAG system using LangChain libraries and FAISS

Chatbot Abilities

- Create/cancel/edit appointments
- Answer health-related queries by making API call to Medline
- Answer clinic operation questions (ie. parking, hours, location, billing etc) by the RAG system

Appointments

- I created an appointments table using SQLite where all of the clinic appointments are stored.
- The local repository and appointments communicate directly with the database, providing all of the required create/search/cancel functions.
- Then, I created an API layer for the agent backend to use and connected it to the LangGraph system by setting the functions as a @tool.

RAG System

- I created dummy documentation for a fictitious clinic with details about hours, parking, location, insurance.
- Then I used the LangChain libraries to perform chunking
- Then, I used OpenAIEmbeddings to vectorize the chunks, then used FAISS to create the vector store
- I created a retriever which then is used a @tool for LangGraph

Screenshots

• Patient view

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Patient View

Doctor View

Select a patient

James Okafor (38, Male)

Search disease-related information

e.g. latest treatments for chronic kidney disease

Clear

Generally healthy adult with mild persistent asthma and seasonal allergies.

Age: 38 · Sex: Male · Active diagnoses: 2

Diagnoses

- Mild persistent asthma — since 2009
- Seasonal allergic rhinitis — since 2015

Medications

- Albuterol inhaler as needed
- Fluticasone nasal spray (seasonal)

Allergies

- Pollen
- Dust mites

17

Appointments

Mon Jun 22, 09:30

Dr. Nguyen · General Physician

test reason

Cancel

Edit

Book appointment

• Doctor view

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Patient View

Doctor View

Select a doctor

Dr. Lee — General Physician

Patient search

Search patients by name

e.g. Robert

Type a patient's name to search.

Medline search

Search disease-related information

e.g. latest treatments for chronic kidney disease

Clear

Dr. Lee — General Physician · Appointments

- 2026-06-23T10:00:00 — Maria Gomez (Post-MI review) · booked

3

- Chatbot assistance - general health query

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[Patient View](#) [Doctor View](#)

Select a patient

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Allergies

Health Assistant

What are some treatment options for my asthma

For asthma treatment, you'll work with your healthcare provider to create a personalized plan, which may include:

- Avoiding Triggers:** Identify and avoid things that trigger your asthma, such as tobacco smoke, pollen, or pet dander.
- Quick-Relief Medicines:** Also known as rescue inhalers, these provide fast relief during an asthma attack by opening the airways.
- Control Medicines:** Taken daily to reduce airway inflammation and prevent symptoms. These are important for long-term control and prevention of asthma attacks.

- Chatbot assistance - clinic operations question

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[Patient View](#) [Doctor View](#)

Select a patient

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e.g. latest treatments for chronic kidney disease

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Allergies

Health Assistant

What are the clinic hours

The clinic is open **Monday to Friday, 9:00 AM to 5:00 PM**. It is closed on weekends and statutory holidays. Appointments are primarily by appointment, and each one lasts **30 minutes**. Same-day slots for urgent concerns are released each morning at 9:00 AM, and walk-ins are accommodated when the schedule allows. If you need to cancel an appointment, please provide at least **24 hours' notice**. Appointments can be booked or canceled through the clinic's online assistant, by phone at (416) 555-0123, or in person at reception.

> Agent trace (2 steps)

- Chatbot assistance - appointment questions

Agentic Healthcare Assistant

[Patient View](#) [Doctor View](#)

Select a patient

James Okafor (38, Male)

Search disease-related information

e.g. latest treatments for chronic kidney disease

Generally healthy adult with mild persistent asthma and seasonal allergies.

Age: 38 · Sex: Male · Active diagnoses: 2

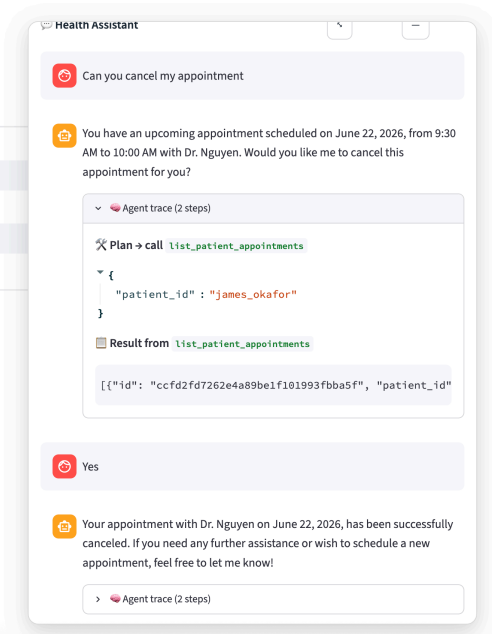
Diagnoses

- Mild persistent asthma — since 2009
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Medications

- Albuterol inhaler as needed
- Fluticasone nasal spray (seasonal)

Allergies



- Doctor view - patient and Medline search

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[Patient View](#) [Doctor View](#)

Select a doctor

Dr. Lee — General Physician

Patient search

Search patients by name

maria gomez

Maria Gomez (54, Female)

Type 2 diabetes with obesity and a prior myocardial infarction.

Age: 54 · Sex: Female · Active diagnoses: 3

⚠ Cardiac history — coordinate any new medication with cardiology.

Diagnoses

- Type 2 diabetes mellitus — since 2012
- Obesity (BMI 34) — since 2010
- Myocardial infarction (post-MI) — since 2022

Medications

- Atorvastatin 40 mg daily
- Aspirin 81 mg daily
- Empagliflozin 10 mg daily

Allergies

- Sulfur dyes

Medline search

Search disease-related information

latest treatments for chronic kidney disease

Clear

Heart Diseases · National Library of Medicine

What is heart disease?Heart disease is a general term that includes many types of heart problems. It's a type of cardiovascular disease, which means heart and blood vessel disease.Heart disease is the leading cause of death in the United States, but there are ways to prevent and manage many types of heart disease.What are the types of heart disease?There are many different types of heart disease. Some you may be born with, called congenital heart disease. Other types develop during your lifetime. Coronary artery disease (also called coronary heart disease) is the most common type of heart disease. It happens slowly over time when a sticky substance called plaque builds up in the arteries that supply your heart muscle with blood. The plaque narrows or blocks blood flow to the heart muscle and can lead to other heart problems: Angina - chest pain from lack of blood flowHeart attacks - when part of the heart muscle dies from loss of blood flowHeart failure - when your heart can't pump enough blood to meet your body's needsArrhythmia - a problem with the rate or rhythm of your heartbeatOther types of heart diseases may affect your heart valves or heart muscle (cardiomyopathy).What causes heart diseases?The causes of heart disease depend on the type of disease. Some possible causes include lifestyle, genetics, infections, medicines, and other diseases. Who is more likely to develop heart diseases?There are many different factors that can make you more likely to develop heart disease. Some of these factors you can change, but others you cannot. Age. Your risk of heart disease goes up as you get older. Sex. Some factors may affect heart disease risk differently in women than in men. Family history and genetics. A family history of early heart disease raises your risk. And research has shown that some genes are linked to a higher risk of certain heart diseases. Race/ethnicity. Heart disease is the leading cause of death in most racial and ethnic groups in the United States. However, certain groups have higher risks than others. This includes Black people, White people, and American Indian/Alaska Native people. Lifestyle habits. Over time, unhealthy lifestyle habits can raise your risk of heart disease. These can include: Eating a diet high in saturated fats, refined carbohydrates, and salt. Not getting enough physical activity. Drinking too much alcohol. Smoking and exposure to secondhand smoke. Too much stress. Having other medical conditions can raise your risk of heart diseases. These conditions include: High blood pressure. High cholesterol levels. Diabetes. Obesity. Autoimmune and inflammatory diseases. Chronic kidney disease. Metabolic syndrome. What are the symptoms of heart disease?Your symptoms will depend on the type of heart disease you have. You may not have symptoms at first. In some cases, you may not know you have heart disease until you have a complication such as a heart attack. How are heart diseases diagnosed?To find out if you have heart disease, your health care provider will ask about your medical history, including your symptoms. Ask about your family health history, including relatives who have had heart disease. A physical exam. Likely run heart tests and blood tests in some cases, your provider may refer you to a cardiologist (a doctor who specializes in heart diseases) for tests, diagnosis, and care. What are the treatments for heart disease?Treatment plans for heart disease depend on the type of heart disease you have, how serious your symptoms are, and what other health conditions you have. Possible treatments may include: Heart-healthy lifestyle changes. Medicines. Procedures or surgeries. Cardiac rehabilitation. Can heart diseases be prevented?You may be able to lower your risk of certain heart diseases by making heart-healthy lifestyle changes and managing any other medical conditions you have. [National Heart, Lung, and Blood Institute](#)

Unit test results

```
You can now view your Streamlit app in your browser.

Local URL: http://localhost:8562
Network URL: http://192.168.1.216:8562

For better performance, install the Watchdog module:

$ xcode-select --install
$ pip install watchdog

^C Stopping...

nour@Mac capstone % make eval
.venv/bin/python -m ai.eval.run_eval

=== Evaluation Report ===

[appointments] success rate: 100% (5/5)
[PASS] find_slots_returns_slots - 16 slots
[PASS] book_creates_appointment - id=00061263c28742e593bf08e5e67b0600
[PASS] booked_slot_removed_from_availability
[PASS] cancel_sets_status_cancelled
[PASS] cancelled_slot_reopened

[semantic] overall accuracy: 100%
clinic_faq: 100%
medline: 100%
[PASS] (clinic_faq) Where can I park at the clinic?
[PASS] (clinic_faq) What are the clinic's opening hours?
[PASS] (clinic_faq) Do you accept insurance?
[PASS] (medline) What is type 2 diabetes?
[PASS] (medline) What are common symptoms of asthma?

Logged results to /Users/nour/Desktop/Nour/Coursework/Applied AI/capstone/ai/eval/results/eval_20260622_161342.j
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nour@Mac capstone % █
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